By TOUCH DEN · November 11, 2026

Passwords and Authentication



Video 1 · Authentication and the Three Factors

Security Fundamentals · Section 6

In this video, you will learn:

- What authentication is
- The three factors of proof
- Why combining factors is strong

What authentication is

Authentication is proving that you are who you say you are. *The lock on the front door of almost every system.*

- ***Identification:*** *claiming an identity ("I am Dara"), a username, not a secret.*
- ***Authentication:*** *proving it ("here is my password / fingerprint / code").*

The username is often public (an email). The whole strength of a login rests on how good the ***proof*** *is.*

The three factors

Factor	Meaning	Examples
Something you know	A secret in your head	Password, PIN
Something you have	A thing you possess	Phone, security key
Something you are	A body trait	Fingerprint, face

Each has a weakness: what you know can be phished; what you have can be stolen; what you are can't be changed if it leaks.

Strength comes from combining _different_ factors.

Recap: authentication and factors

- *Authentication proves who you are; the username is just the claim.*
- *Three factors: something you know, have, are.*
- *Two different factors are far stronger than two of the same kind.*

Next video: *passwords, why they fail, and how to make them strong.*

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Video 2 · Passwords: Why They Fail, and How to Make Them Strong

Security Fundamentals · Section 6

In this video, you will learn:

- Why passwords fail so often
- How passwords are stored and cracked
- Strong passwords, passphrases, and password managers

Why passwords fail

- *People choose **weak** ones ("123456", "password").*
- *People **reuse** them, so one breach unlocks many accounts.*
- *People can be **tricked** into giving them away.*
- *They can be **guessed in bulk**, billions of tries against a stolen file.*
- *We simply have **too many** to manage by memory.*

A password alone is weak. It must be made strong, and backed by a second factor.

How passwords are stored and cracked

A system must store something to check against, and stores get stolen.

- **Plain text (worst):** a leak exposes every password instantly.
- **Hashing (right):** store a one-way "fingerprint"; you can't reverse it.

Attackers crack stolen hashes **offline**, billions of guesses a second. Two defenses:

- **Salting:** a unique random value per password, so identical passwords differ.
- **Slow hashing** (bcrypt, Argon2): each guess costs real time.

Length helps: every extra character multiplies the guessing work.

Strong passwords and password managers

Length and unpredictability beat strange symbols.

- ***Length wins:*** *aim for 12–16+ characters.*
- ***Passphrases:*** *4–5 random words,* `correct-harbor-tiger-bottle` *, long, memorable, hard to guess.*
- `P@ssw0rd!` ***is weak:*** *the substitutions are the first thing tools try.*
- ***Unique per site****, always.*

A ***password manager*** *makes unique strong passwords effortless, fills them* ***only on the correct site*** *(anti-phishing), and stores them* ***encrypted****. One of the best habits you can adopt.*

Recap: passwords

- *Passwords fail because they're weak, reused, phishable, guessable, and too many.*
- *Good storage is salted, slow hashing, never plain text; length multiplies the work.*
- *Long unique passphrases + a password manager solve most of it.*

Next video: *the two steps that matter most, MFA and passkeys.*

Passwords and Authentication



Video 3 · MFA and Passkeys

Security Fundamentals · Section 6

In this video, you will learn:

- What multi-factor authentication is, and why it works
- Which second factors are strongest
- What passkeys are, and why they're the future

Multi-factor authentication

MFA requires two _different_ factors: *usually your password (know) + a code or tap on your phone (have).*

The single most effective step *for most people and businesses: a stolen password is now useless on its own.*

- *Credential stuffing, most phishing, and bulk cracking all **fail** against MFA.*

*Not all equal: **SMS** is weakest (can be redirected); **authenticator apps** stronger; **security keys / passkeys** strongest (also verify the site).*

Turn it on everywhere, email first. *Never approve a prompt you didn't start.*

Passkeys: the direction of travel

*A **passkey** lives on your device and proves who you are **without sending a password**. You unlock it with your fingerprint, face, or PIN.*

- ***Nothing to phish:** no password to type; it checks it's the real site.*
- ***Nothing to reuse or leak:** unique per site; no stored secret to crack.*
- ***Easy:** signing in is just unlocking your device.*

Built on public-key cryptography, supported by the major platforms, and growing.

Recap: MFA and passkeys

- **MFA** adds a second, different factor and defeats stolen passwords; turn it on everywhere, email first.
- SMS is the weakest factor; apps and keys are stronger.
- **Passkeys** remove the password: unphishable, unique per site, the future.
- **Practical plan:** password manager + MFA everywhere + passkeys where offered.

Next lesson: once a system knows who you are, how it decides what you may do, access control.